

RomBuzz Meet in the Middle

Backend file reference - clean MongoDB + Geoapify rebuild

Generated: 2026-05-25 18:12 UTC

Purpose

This PDF lists every new or updated backend file used for the clean Meet in the Middle rebuild, plus the old legacy files that should not be reused. Keep this as a debugging handoff: later, when mobile or web work starts, upload only the latest relevant files from the sections below instead of dumping the whole backend.

Core decisions already made

Decision	Status
Storage	MongoDB only for this feature. Do not use LowDB for sessions, locations, or meet state.
Provider	Geoapify for the first production-ready version. Provider wrapper stays modular for future providers.
Security	Geoapify key lives only in Render backend env as GEOAPIFY_API_KEY. Mobile must never receive this key.
Socket names	Use meetMiddle:* only. Do not use old meet:* names because old broken handlers already exist.
Backend-first design	Backend is mobile-first now, but generic enough for future web reuse.

New files created for clean Meet in the Middle

File	Purpose	Upload later when debugging...
utils/geo.js	Shared coordinate validation, geographic midpoint calculation, distance helpers, meter/mile helpers, and number clamping.	Midpoint math, invalid coordinates, wrong distances, or radius behavior.
services/geoapifyService.js	Backend-only Geoapify wrapper. Calls /v2/places, normalizes provider payloads into clean RomBuzz place objects, expands radius, exposes safe provider health status.	Places, categories, empty results, Geoapify errors, bad addresses, missing coords, provider health, or env config.
models/MeetMiddleSession.js	MongoDB/Mongoose model for temporary meet sessions, users, coordsByUser, midpoint, places, selected place, status, timestamps, and TTL cleanup.	Session persistence, session expiration, selected place state, status transitions, or Mongo records.
services/meetMiddleService.js	Central business logic. Validates users/matches, creates sessions, stores user coords, calculates midpoint, calls Geoapify, handles select/accept/reject/cancel/complete.	Most backend logic bugs. Upload this first for almost any meet flow issue.

File	Purpose	Upload later when debugging...
routes/meetMiddle.js	Protected HTTP API layer mounted at /api/meet-middle. Includes health, provider-health, protected provider-smoke, request, location, decline, place select/accept/reject, cancel, complete.	REST endpoint issues, route mounting, auth behavior, or Render endpoint issues.
sockets/meetMiddleSocket.js	Clean realtime Socket.IO layer using meetMiddle:* events only. Calls service methods and emits request/location/suggestions/place/final/cancel/complete events.	Realtime app issue, missing popup, duplicate events, no final confirmation, or socket errors.
services/meetMiddleChatService.js	Creates a MongoDB ChatRoom system message when a meetup place is fully confirmed. Keeps chat-message creation out of sockets.	Confirmed meetup chat bubble, direct:message, chat:message, roomId, or saved chat history.
services/meetMiddleRateLimitService.js	MongoDB-backed cooldown for meet requests. Protects HTTP and socket flows because it runs inside the central service.	Cooldown, spam prevention, retryInMs, or 429 MEET_REQUEST_COOLDOWN issue.

Existing files updated

File	What changed	Why it matters
index.js	Added <code>app.use("/api/meet-middle", require("./routes/meetMiddle"));</code> and registered <code>registerMeetMiddleSockets(io)</code> . Old <code>registerMeetSockets(io)</code> was left in place for now.	Exposes the new REST API and new non-conflicting socket layer without breaking old app code still using legacy <code>meet:*</code> events.
services/geoapifyService.js	Added <code>getGeoapifyHealthStatus()</code> , exported <code>searchPlacesAroundPoint</code> , and removed/commented <code>apiKeyPreview</code> after verification.	Lets Render confirm provider env is configured without exposing the actual key. <code>searchPlacesAroundPoint</code> powers smoke tests and place lookup.
routes/meetMiddle.js	Added <code>/health</code> , <code>/provider-health</code> , <code>/provider-smoke</code> , then protected <code>/provider-smoke</code> with <code>authMiddleware</code> .	Allows safe Render verification while preventing public quota abuse through <code>provider-smoke</code> .
services/meetMiddleService.js	Added <code>enforceMeetMiddleRequestCooldown()</code> inside <code>createMeetRequest()</code> .	Protects both HTTP and socket meet requests from spam.
sockets/meetMiddleSocket.js	Added <code>meetMiddleChatService</code> integration inside <code>meetMiddle:place:accept</code> , creating <code>chat:message</code> and <code>direct:message</code> after final confirmation.	Confirmed meetup now creates a persistent chat bubble for both users.

Legacy/old files - do not build the new feature from these

Old file	Current status	Rule
routes/meet.js	Legacy <code>safe-meet/old meet</code> route. Still mounted under <code>/api</code> for now.	Do not reuse for this rebuild. Later remove only after mobile/web no longer depend on it.
sockets/meetSocket.js	Legacy <code>Meet-in-Middle</code> socket file with old <code>meet:*</code> events and inconsistent behavior.	Do not modify for the new feature except eventual cleanup. New app must use <code>meetMiddle:*</code> only.
sockets/connection.js	Main legacy connection handler; contains old <code>meet:request</code> / <code>meet:accept</code> logic and <code>LowDB</code> patterns.	Do not add new <code>Meet</code> logic here. Avoid old <code>meet:*</code> events.
models/db.lowdb and any <code>LowDB</code> write path	Legacy backend storage still used elsewhere in <code>RomBuzz</code> , but not for this feature.	Never use <code>LowDB</code> for <code>Meet</code> in the <code>Middle</code> sessions, coords, places, or state.
Any old <code>/api/meet-suggest</code> or <code>/api/meet</code> flow	Old <code>self-fetch/provider</code> pattern from previous implementation.	Do not call backend from backend for suggestions. Use services directly.

New HTTP endpoints

Endpoint	Auth	Purpose
GET /api/meet-middle/health	Public	Confirms route is mounted on Render.
GET /api/meet-middle/provider-health	Public	Confirms Geoapify env is configured without exposing key.
GET /api/meet-middle/provider-smoke	Protected	Calls Geoapify using fixed coords. Protected to avoid quota abuse.
POST /api/meet-middle/request	Protected	Creates/reuses a meet request session for matched users.
POST /api/meet-middle/:sessionId/location	Protected	Stores caller coords in session; when both shared, calculates midpoint and fetches places.
POST /api/meet-middle/:sessionId/decline	Protected	Declines a meet request.
POST /api/meet-middle/:sessionId/place/select	Protected	One user selects a place and asks the other to confirm.
POST /api/meet-middle/:sessionId/place/accept	Protected	Other user accepts selected place and finalizes session.
POST /api/meet-middle/:sessionId/place/reject	Protected	Other user rejects selected place; users remain in browsing flow.
POST /api/meet-middle/:sessionId/cancel	Protected	Cancels a session.
POST /api/meet-middle/:sessionId/complete	Protected	Marks a confirmed meetup complete.

New Socket.IO event namespace

The new socket system must use only `meetMiddle:*` events. Do not use old `meet:*` events because old legacy handlers still exist in the backend.

Client emits	Server emits / result
<code>meetMiddle:request</code>	<code>meetMiddle:request:sent</code> to sender; <code>meetMiddle:request:received</code> to receiver.
<code>meetMiddle:location:share</code>	<code>meetMiddle:location:waiting</code> or <code>meetMiddle:location:peer-shared</code> until both users share; then <code>meetMiddle:suggestions:ready</code> to both.
<code>meetMiddle:decline</code>	<code>meetMiddle:declined</code> to both users.
<code>meetMiddle:place:select</code>	<code>meetMiddle:place:selected</code> to selector; <code>meetMiddle:place:confirmation-needed</code> to peer.
<code>meetMiddle:place:accept</code>	<code>meetMiddle:final-confirmed</code> to both; creates <code>chat:message</code> and <code>direct:message</code> .
<code>meetMiddle:place:reject</code>	<code>meetMiddle:place:rejected</code> to both.
<code>meetMiddle:cancel</code>	<code>meetMiddle:cancelled</code> to both.
<code>meetMiddle:complete</code>	<code>meetMiddle:completed</code> to both.
Any socket error	<code>meetMiddle:error</code> with event, error, message, <code>retryInMs</code> , and optional <code>sessionId</code> .

Render environment variables used

Variable	Purpose
<code>GEOAPIFY_API_KEY</code>	Secret Geoapify backend key. Store only in Render. Never commit and never put in Expo.
<code>GEOAPIFY_BASE_URL</code>	Usually <code>https://api.geoapify.com</code> .
<code>MEET_MIDDLE_CACHE_TTL_SECONDS</code>	Planned cache TTL setting, currently exposed in provider health for config visibility.
<code>MEET_MIDDLE_DEFAULT_RADIUS_METERS</code>	Default first search radius. Current plan: 2000 meters.
<code>MEET_MIDDLE_MAX_RADIUS_METERS</code>	Maximum expanded search radius. Current plan: 32000 meters.
<code>MEET_MIDDLE_MAX_RESULTS</code>	Max place results returned by provider wrapper. Current plan: 20.
<code>MEET_MIDDLE_REQUEST_COOLDOWN_SECONDS</code>	Request spam cooldown. Current plan: 30 seconds.
<code>MEET_MIDDLE_SESSION_TTL_MINUTES</code>	Meet session expiration. Default in service: 45 minutes.

What to upload later for debugging

Problem type	Upload these latest files first
Backend place search or Geoapify bug	<code>services/geoapifyService.js</code> , <code>utils/geo.js</code> , <code>routes/meetMiddle.js</code>
Session/status/coords bug	<code>services/meetMiddleService.js</code> , <code>models/MeetMiddleSession.js</code> , <code>models/Match.js</code>

Problem type	Upload these latest files first
Socket popup/realtime bug	sockets/meetMiddleSocket.js, services/meetMiddleService.js, index.js, current mobile socket file
Confirmed meetup chat bubble bug	services/meetMiddleChatService.js, sockets/meetMiddleSocket.js, models/ChatRoom.js
Cooldown/spam bug	services/meetMiddleRateLimitService.js, services/meetMiddleService.js
Route not live on Render	index.js, routes/meetMiddle.js, Render logs
Future web integration	routes/meetMiddle.js, services/meetMiddleService.js, services/geoapifyService.js, sockets/meetMiddleSocket.js, utils/geo.js

Known current state

Backend foundation is ready for mobile integration. The old legacy meet route/socket code still exists, but the new implementation intentionally avoids conflicts through /api/meet-middle routes and meetMiddle:* socket events. The next mobile step should be a small placeholder route/button from chat before building the full map UI.

Important cleanup before public launch

Cleanup item	Why
Remove or admin-protect provider-health if desired	It does not expose the key, but public operational metadata should be minimal before launch.
Keep provider-smoke protected or remove it	It calls Geoapify and can burn quota if public.
Delete old meet:* client usage after mobile/web migration	Prevents accidental use of broken legacy handlers.
Eventually remove legacy routes/meet.js and sockets/meetSocket.js	Only after confirming no app/client code depends on them.